# FAHAD MOLLAH

AI / Machine Learning Student | Applied AI, Data & Autonomous Systems

Ingolstadt, Germany | Eligible to work up to 20h/week (student visa) | fahadmollah43@gmail.com | +49 178 6279412 | github.com/Fahad4969 | linkedin.com/in/fahad6949 | fahadmollah.netlify.app

## PROFILE

AI Engineering master's student at THI Ingolstadt with a genuine interest in systems that can sense, decide, and act on their own, and in the discipline of proving they actually work. That's shown up in a few different ways: training reinforcement learning agents from scratch and measuring, rigorously, whether they were really improving; programming a robotic arm's real-time perception and control; and, before the master's, spending two years in enterprise IT turning messy operational data into reports people trusted enough to act on. I move comfortably between the ML side of a problem and the practical software engineering around it, and I'm just as happy working independently as pairing with a team. Currently looking for a Werkstudent, internship, or thesis opportunity in AI, machine learning, or autonomous systems.

## TECHNICAL SKILLS

- **Programming:** Python, C++, SQL, JavaScript
- **ML / Data:** PyTorch, TensorFlow (basic), OpenCV, pandas, NumPy, scikit-learn, feature engineering, EDA, data pipelines & validation
- **AI Foundations:** Theoretical foundations of LLM / Transformer architectures; genuine interest in agent-based AI systems and automation logic
- **Autonomous Systems:** Sensor data fusion (camera, radar, LiDAR), system identification & simulation, edge computing architectures, safety-critical AI systems
- **Tools:** Git, Docker (basic), Jupyter, Linux/CLI, VS Code, MS Excel
- **Languages:** English (B2), German (A1, actively improving), Bangla (native)
- **Soft Skills:** Analytical thinking, problem-solving, independent and collaborative work, clear communication

## PROJECTS

### Reinforcement Learning Training Pipeline for Autonomous Inspection
*03/2025 – 07/2025*

*Python, Pygame, Q-learning, DQN*

- Built a custom simulation environment and trained Q-learning and DQN agents in it, using experience replay, reward shaping, and epsilon-greedy decay to help them learn efficiently.
- Set up a reproducible pipeline with fixed seeds and structured configs, so different agent variants could be compared fairly against each other.
- Tracked convergence, success rate, and episode length throughout training, and used those numbers to guide tuning and iron out instability.

### Autonomous Object Sorting System (Robotics)
*03/2025 – 07/2025*

*Arduino, Braccio, OpenCV, Motion Control*

- Programmed pick-and-place routines for a Braccio robotic arm, then added a computer-vision-based color-detection module (OpenCV) so it could classify and sort objects on its own.
- Folded the arm into a multi-robot workflow and ran it under continuous operation to see how reliably it held up.

## EXPERIENCE

### IT Executive
*07/2022 – 06/2024*

*Technohaven Company Limited, Dhaka*

- Dug into operational data from enterprise systems to spot trends and support process-improvement decisions across teams.
- Checked data availability and quality before it went into internal reports, catching gaps before they reached stakeholders.
- Put together structured reports and presentations that turned raw operational data into something decision-makers could actually use.

### IT Intern
*02/2022 – 04/2022*

*American International University, Dhaka*

- Kept university IT systems and network infrastructure running, troubleshooting issues as they came up.
- Logged recurring technical issues and their fixes, building a reference that cut down repeat-issue response time.

## EDUCATION

### M.Eng. AI Engineering of Autonomous Systems
*03/2024 – Present*

*Technische Hochschule Ingolstadt, Germany*

- Focus on machine learning, computer vision, and robotics, with reinforcement learning (Q-learning, DQN) explored hands-on through independent projects.

### B.Sc. Software Engineering
*01/2018 – 06/2022*

*American International University, Bangladesh*